PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR NORTH WESTERN ZONE MOCK EXAMINATION 2024

PHYSICS 2A

TIME 2:30 HRS **2024**

INSTRUCTIONS

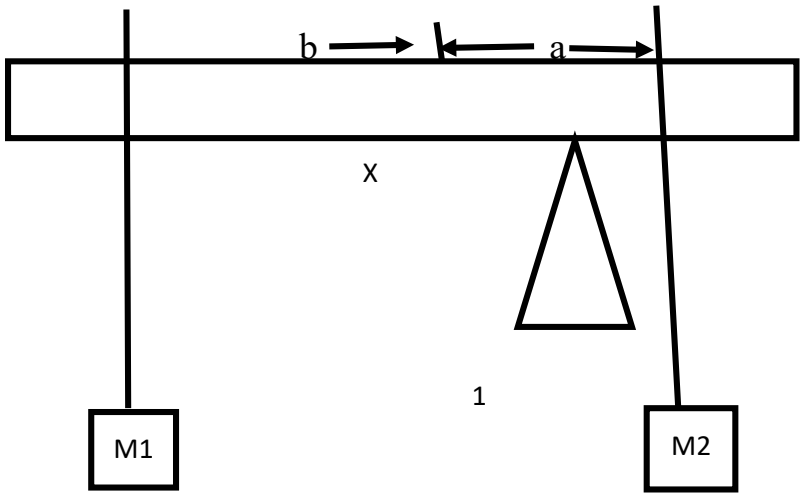
(a) This paper consist of a total of two questions
(b) Answer all questions
(c) Non – programmable calculators and mathematical tables may be used
(d) The following constants may be used
(i) Acceleration due to gravity, g 10m/s^2
(ii) Pie = 3.14

01. The aim of this experiment is to determine unknown mass m_2 and the density of a meter rule, You are supplied with the following; standard mass (M_1) of 50g, unknown mass M_2, a meter rule, a knife edge and two pieces of cotton thread.

Procedures

(a) Balance the meter rule on the knife edge then mark its center of gravity as X
(b) Place the knife edge 5cm from X on the right – hand side
(c) Hang the mass M_2 at a distance 'a' of 10cm from the fulcrum (knife-edge) on right – hand side of the fulcrum so as to balance the meter rule horizontally as shown in figure below
(d) Read and record the balancing distance 'b' of M1 from the fulcrum.
(e) Repeat the procedures in (c)and (d) above by varying the distance "a" and recording the balancing distance "b" as shown in the table below.

NB: Always the position of the fulcrum should be kept constant (i) Table of result



Distance “a” in cm	10	12	14	16	18
Distance “b” in cm					

(ii) Plot the graph of ‘a’ (in vertical axis) against “b” (in horizontal axis)

(iii) Determine the slope of the graph (G) and the intercept K on the “a” axis

(iv) Use the slope G, the intercept K and the principle concerned to determine the mass of the meter rule and the unknown mass M_2

(v) Use a veneer caliper to obtain the breath and the thickness of the meter rule then calculate its volume “V”

(vi) Using the result obtained in (iv) and (v) above to determine the density of the meter rule.

2. The aim of this experiment is to determine the internal resistance r and the EMF E of the dry cell D.

(i) You are provided with a dry cell, an ammeter, voltmeter, key and resistance box

(ii) Connect the apparatus so that you can get reading of ammeter and corresponding reading of voltmeter which was to be connected parallel to the dry cell labelled D

(iii) Set the resistance box to be 152, read and record the ammeter and voltmeter readings.

(iv) Repeat the procedure for resistance R = 2,3,4 and 5 ohms

(v) Draw a circuit diagram of arrangement used

(vi) Draw a graph V on y – axis against I on X – axis

(vii) Write down the equation represented by the graph in terms of E,R,V and I

(viii) From your graph, find;

(a) The values of electromotive force E

(b) The internal resistance r

(c) The maximum current the cell can deliver to the circuit